SHIP TYPES 1.2 · ABOUT 8 MIN

Deadweight, Volume and Linear-Dimension Carriers

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Stowage Factor and Capacity Factor

A PROPERTY OF THE CARGO

$$\mathrm{SF} = \frac{V_{\mathrm{cargo}}}{W} \quad \left[\mathrm{m}^3/\mathrm{t}\right]$$

Low: ore, coal, grain, oil · High: cotton, fruit, cars

A PROPERTY OF THE SHIP

$$\mathrm{CF} = \frac{V}{\mathrm{DWT}} \quad \left[\mathrm{m}^3/\mathrm{t}\right]$$

Set by the designer, through depth, beam and hold shape

Goal: when loading a cargo of weight **W** (t), find what runs out first: the deadweight for cargo **DWc** (t) or the hold volume **V** (m³). The **switch-over SF*** is the stowage factor at which both run out together: a full deadweight of cargo exactly fills the holds.

1 · Weight limit
no heavier than the deadweight for cargo

$$W \le \mathrm{DW}_c$$

2 · Volume limit
W tonnes take SF · W cubic metres

$$\mathrm{SF} \cdot W \le V \quad \Rightarrow \quad W \le \frac{V}{\mathrm{SF}}$$

3 · Both apply
the smaller limit decides

$$W = \min\left(\mathrm{DW}_c\,,\ \frac{V}{\mathrm{SF}}\right)$$

4 · Switch-over
both run out together

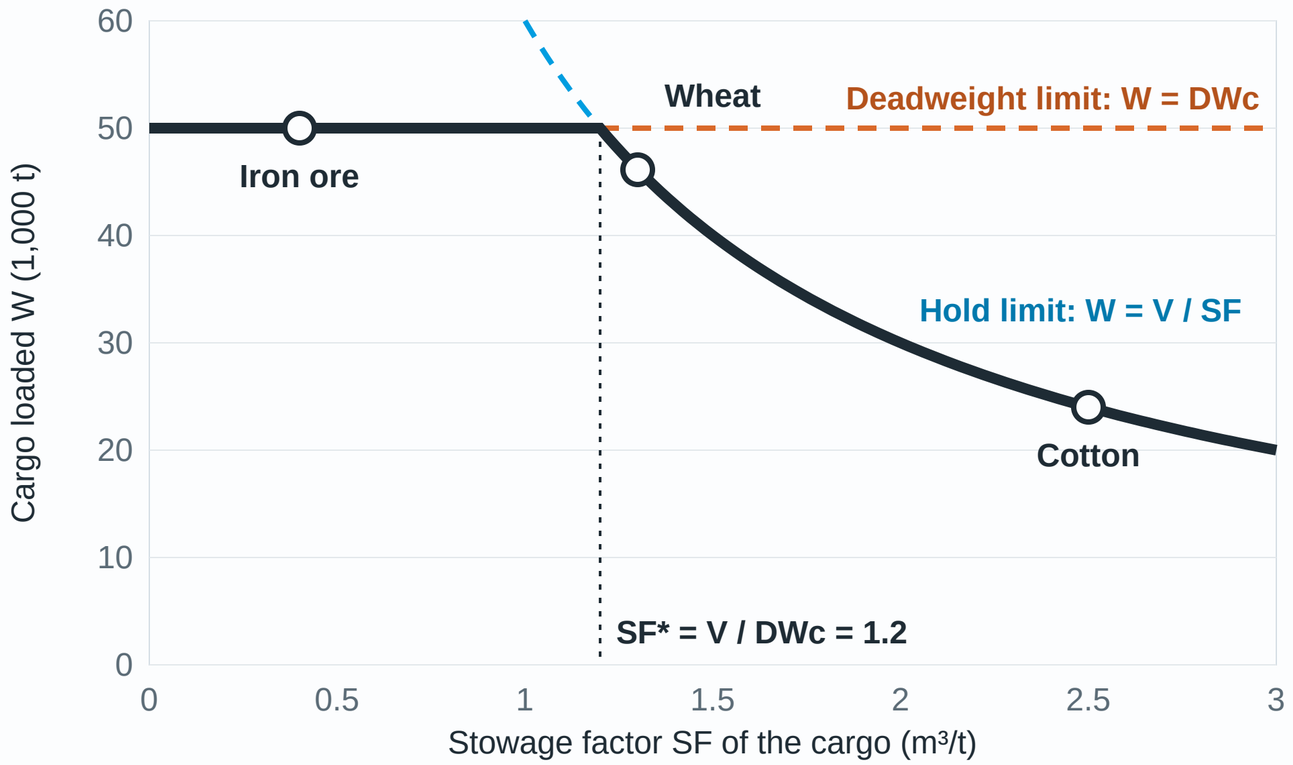
$$\mathrm{DW}_c = \frac{V}{\mathrm{SF}^*} \quad \Rightarrow \quad \mathrm{SF}^* = \frac{V}{\mathrm{DW}_c}$$

5 · In ship terms
k < 1: fuel, water, stores

$$\mathrm{DW}_c = k \cdot \mathrm{DWT}, \quad \mathrm{CF} = \frac{V}{\mathrm{DWT}} \quad \Rightarrow \quad \mathrm{SF}^* = \frac{\mathrm{CF}}{k} > \mathrm{CF}$$

So: SF < SF* → the deadweight runs out first (deadweight carrier); SF > SF* → the holds fill first (volume carrier).

One Ship, Three Cargoes



The ship

Cargo volume V = 60,000 m³
Deadweight for cargo DWc = 50,000 t
Switch-over SF* = 60,000 / 50,000 = **1.2 m³/t**

Cargo	SF	W, t	What runs out
Iron ore	0.4	50,000	Deadweight
Wheat	1.3	46,200	Holds
Cotton	2.5	24,000	Holds

W = min(50,000, 60,000 / SF). Round numbers; SF values are typical.

What Runs Out First When You Load the Ship

Deadweight carriers

Heavy cargo, SF below about 1.3 m³/t. The load line is reached with hold space left. CF below about 1.5 m³/t DWT.

Bulk and ore carriers, crude tankers, tramp cargo ships

Volume carriers

Light cargo, SF above about 2.0 m³/t. The holds are full before the load line; excess freeboard. CF above about 2.5 m³/t DWT.

Ro-Ro, car carriers, Ro-Pax, container ships, reefers, cruise ships

Linear-dimension ships

One of L, B, T or D is fixed by a canal, a port, a river or the size of the cargo unit.

Panamax ships, Great Lakes bulk carriers, cellular container ships

Special-purpose ships

Sized by a mission: bollard pull, ice class, fish hold, deck work. No general method.

Tugs, icebreakers, fishing and offshore vessels, fast craft

Cargo between 1.3 and 2.0 m³/t, or cargoes that vary a lot: the ship can be designed either way.

When One Dimension Is Fixed from Outside

Constraint	Largest ship allowed	Fixes
Panama Canal, old locks (Panamax)	LOA 294.13 m · B 32.31 m · T 12.04 m	Beam, draught
Panama Canal, new locks (Neopanamax)	LOA 366 m · B 51.25 m · T 15.2 m	Beam, draught
St Lawrence Seaway (Seawaymax)	LOA 225.5 m · B 23.8 m · T 8.08 m	Beam, length
Azores island ports	Draught and berth length, each port	To be checked, port by port
ISO container (ISO 668)	8 ft = 2.438 m wide; 8 ft 6 in standard, 9 ft 6 in high-cube	Beam and depth, in steps

> The cost: dimensions jump in steps and the proportions suffer. A 3rd-generation Panamax container ship of about 3,700 TEU: B = 32.2 m, L = 245 m, **L/B = 7.6**.

Which Category Is the Reference Ship?

- Is it the weight of the cargo or the tank volume that runs out first?
- Which line of the brief fixes a main dimension from outside?
- Do the ports on the route limit draught or length?
- So: does the design start from deadweight or from volume?